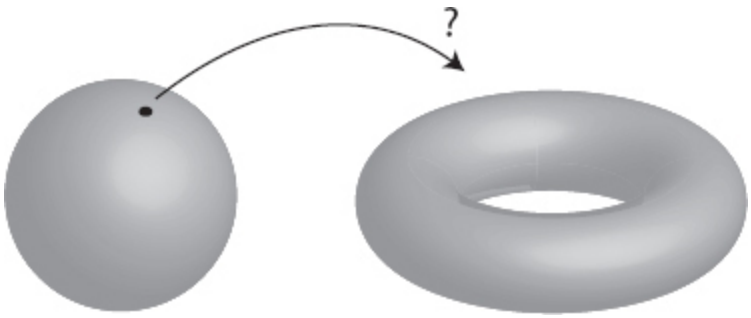
Consider a two-dimensional sphere and torus. Imagine that a friend of yours picks out a point on a sphere, and then asks you to pick out “the same” point on the torus. You’d be stymied, and for good reason; there’s no way to do it.



Apparently, spacetime can’t play the same central role in quantum gravity that it does in the rest of physics. There isn’t a single spacetime, there’s a superposition of many different spacetime geometries. We can’t ask what the probability might be to find an electron at a certain point in space, since there’s no objective way to specify which point we’re talking about.

Quantum gravity, then, comes with a set of conceptual issues that distinguish it from other quantum-mechanical theories. These issues can have important ramifications for the nature of our universe, including the question of what happened at the beginning, or if there was a beginning at all. We can even ask whether space and time are themselves fundamental, or if they emerge out of something deeper.

Just like the foundations of quantum mechanics, the field of quantum gravity was relatively ignored for decades as physicists concentrated on other things. Not completely; Hugh Everett was inspired to propose the Many-Worlds approach in part by thinking about the quantum theory of the entire universe, where gravity plays an important role, and his mentor, John Wheeler, worried about the problem for years. But even putting aside the conceptual issues, other obstacles got in the way of making serious progress on quantizing gravity.